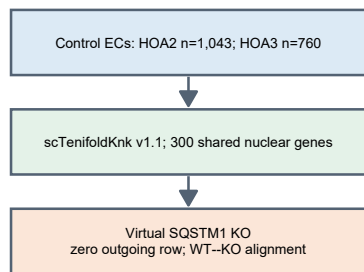


A

Prespecified, donor-separated virtual-knockout design

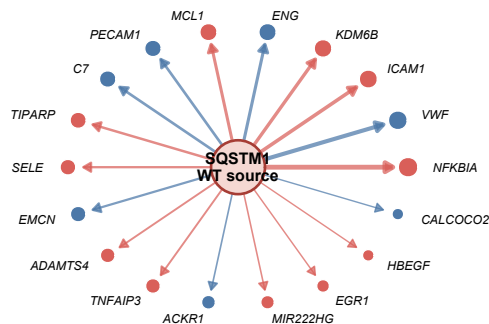


Primary: 20-network $q=0.95$ profile; sensitivity: package-default profile

B

Top nuclear WT outgoing edges

Mean absolute edge weight across HOA2 and HOA3; topology is descriptive

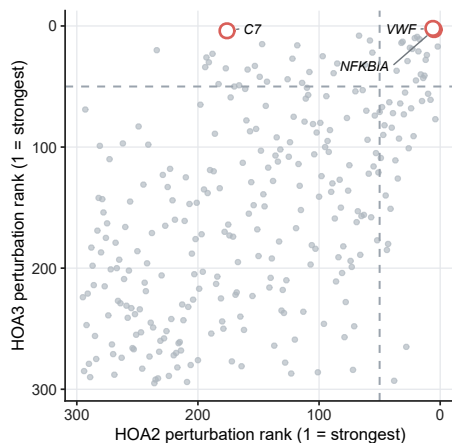


Edge sign → Negative → Positive

C

Cross-donor nuclear rank agreement

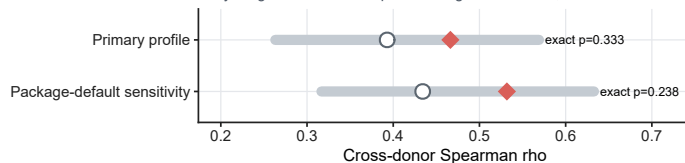
Donor-correlated but non-specific; outlines: HOA3 FDR<0.05



D

Matched-gene calibration

Grey range and circle: comparator range and median; diamond: SQSTM1



E

Matched-null pathway rank recovery

* empirical FDR<0.05 in one donor only; no pathway replicated

